Name of Student: Minisha Singh	**PRN: 25070122292**
Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 21	
TITLE : JDBC DriverManager, Connection, and Statement	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives

To understand and use the core JDBC classes for database connectivity.

Exercise

1. **Create a program to connect Java with database and display database connection status.**

Code:

```
import java.sql.*;

public class DatabaseConnection {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "mini@12";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            System.out.println("Database connected successfully.");

            con.close();

        } catch (Exception e) {
```

```
            System.out.println("Database connection failed.");
            System.out.println(e);
        }
    }
}
```

Output:

```
PS C:\Users\minis\Downloads\javaprograms>  c:; cd
'c:\Users\minis\Downloads\javaprograms'; & 'C:\Pro
gram Files\Java\jdk-27\bin\java.exe' '-agentlib:jd
wp=transport=dt_socket,server=n,suspend=y,address=
localhost:58986' '@C:\Users\minis\AppData\Local\Te
mp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'Database
Connection'
Database connected successfully.
PS C:\Users\minis\Downloads\javaprograms>
```

2. **Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.**

Code:

```
import java.sql.*;

public class StudentDatabaseConnection {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "mini@12";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            System.out.println("Student database connected successfully.");

            con.close();

        } catch (Exception e) {
```

```
            System.out.println("Student database connection failed.");
            System.out.println(e);
        }
    }
}
```

Output:

```
PS C:\Users\minis\Downloads\javaprograms>  & 'C:\P
rogram Files\Java\jdk-27\bin\java.exe' '-agentlib:
jdwp=transport=dt_socket,server=n,suspend=y,addres
s=localhost:59423' '@C:\Users\minis\AppData\Local\
Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'Studen
tDatabaseConnection'
Student database connected successfully.
PS C:\Users\minis\Downloads\javaprograms>
```

Conclusion

In this practical, we learned how to connect a Java program with a MySQL database using JDBC. We established the database connection and displayed the connection status after successful connectivity. We also understood how Java communicates with the database using the JDBC driver and connection methods. The Student database connection was successfully established and verified through the output message.